

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0377

Scott Branch 138 kV
October 28, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0377. The Interconnection Customer proposes a 277.4 MW solar photovoltaic facility interconnecting at the Scott Branch 138 kV substation in the Kentucky area. The request was received on September 07, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 229 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$30,310,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Ioco-Kean Mtn And Grassy Crk 138.0 kV
Contingency: Clinch Riv-Carlton Hills 345.0 kV Line
Loading: 239.9% (Rating: 256 MVA)
DFAX: 6.7%
Pre-project loading: 232.7%
2. Facility: Kean Mtn And Grassy Creek 345/1 kV Autotransformer
Contingency: Clinch Riv-Carlton Hills 345.0 kV Line
Loading: 171.9% (Rating: 471 MVA)
DFAX: 12.9%
Pre-project loading: 164.3%
3. Facility: Kean Mtn And Grassy Creek 138/1kV Transformer Bank
Contingency: Clinch Riv-Carlton Hills 345.0 kV
Loading: 171.9% (Rating: 471 MVA)
DFAX: 12.9%
Pre-project loading: 164.3%
4. Facility: Tloga-Parke 138.0 kV line
Contingency: New Hope Ap-Barr Lake 345.0 kV Line
Loading: 155.5% (Rating: 270 MVA)
DFAX: 7.6%
Pre-project loading: 147.7%
5. Facility: Scott Branch-Sidney Coal 138.0 kV
Contingency: Wansley Combined Cycle-Massapequa 138.0 kV L
Loading: 144.6% (Rating: 289 MVA)
DFAX: 81.1%
Pre-project loading: 66.7%

6. Facility: Wansley Combined Cycle-Massapequa Substation 138.0 kV L
Contingency: Sidney Coal-Scott Branch 138.0 kV Line
Loading: 138.6% (Rating: 274 MVA)
DFAX: 76.2%
Pre-project loading: 61.5%
7. Facility: Leesburg Picciola-Orovada 138.0 kV Line
Contingency: Kean Mtn And Grassy Crk 138-1 kV XFMR
Loading: 137.4% (Rating: 247 MVA)
DFAX: 8.1%
Pre-project loading: 128.2%
8. Facility: Hinton Ap-Antelope Vly Sta Substation 138.0 kV Line
Contingency: Kean Mtn And Grassy Creek-Barr Lake 345.0 kV Line
Loading: 125.5% (Rating: 391 MVA)
DFAX: 5.5%
Pre-project loading: 121.5%
9. Facility: Parke 138/1 kV Autotransformer
Contingency: Barr Lake-New Hope Ap Substation 345.0 kV
Loading: 124.1% (Rating: 338 MVA)
DFAX: 7.6%
Pre-project loading: 117.9%
10. Facility: Sta 11-Stock Island 138.0 kV
Contingency: Kean Mtn And Grassy Crk-Barr Lake 345.0 kV L
Loading: 121.2% (Rating: 242 MVA)
DFAX: 5.7%
Pre-project loading: 114.6%
11. Facility: Springbank-Station 11 138.0 kV
Contingency: Barr Lake-Kean Mtn And Grassy Creek 345.0 kV
Loading: 118.2% (Rating: 234 MVA)
DFAX: 5.7%
Pre-project loading: 111.5%
12. Facility: Optical Coating Lab-Minnix Mountain 138.0 kV
Contingency: Kean Mtn And Grassy Crk-Barr Lake 345.0 kV Line
Loading: 117.2% (Rating: 194 MVA)
DFAX: 5.1%
Pre-project loading: 109.9%
13. Facility: Lovely-Central Ephrata 138.0 kV
Contingency: Kean Mtn And Grassy Crk-Barr Lake 345.0 kV Line
Loading: 115.3% (Rating: 211 MVA)
DFAX: 26.8%
Pre-project loading: 79.9%
14. Facility: Recway-Minnix Mountain 138.0 kV line
Contingency: Kean Mtn And Grassy Creek-Barr Lake 345.0 kV
Loading: 114.3% (Rating: 202 MVA)
DFAX: 5.1%
Pre-project loading: 107.3%
15. Facility: Recway-Prairie City 138.0 kV
Contingency: Kean Mtn And Grassy Creek-Barr Lake 345.0 kV
Loading: 112.6% (Rating: 208 MVA)
DFAX: 5.1%
Pre-project loading: 105.8%
16. Facility: Tortilla-Lovely 138.0 kV
Contingency: Kean Mtn And Grassy Crk-Barr Lake 345.0 kV

- Loading: 112.6% (Rating: 222 MVA)
DFAX: 26.8%
Pre-project loading: 79.0%
17. Facility: Sidney Coal-Bandon 138.0 kV L
Contingency: Kean Mtn And Grassy Creek-Barr Lake 345.0 kV
Loading: 111.4% (Rating: 240 MVA)
DFAX: 26.8%
Pre-project loading: 80.3%
18. Facility: Bandon-Tortilla 138.0 kV
Contingency: Kean Mtn And Grassy Crk-Barr Lake 345.0 kV
Loading: 111.2% (Rating: 229 MVA)
DFAX: 26.8%
Pre-project loading: 78.7%
19. Facility: Lonesome Pine-Municipal Light 138.0 kV
Contingency: Kean Mtn And Grassy Crk-Barr Lake 345.0 kV line
Loading: 109.4% (Rating: 247 MVA)
DFAX: 5.1%
Pre-project loading: 103.6%
20. Facility: Municipal Light-Daffin Gin 138.0 kV Line
Contingency: Kean Mtn And Grassy Creek-Barr Lake Substation 345.0 kV
Loading: 108.7% (Rating: 255 MVA)
DFAX: 5.1%
Pre-project loading: 103.1%
21. Facility: Lonesome Pine-Prairie Ciy 138.0 kV
Contingency: Kean Mtn And Grassy Crk-Barr Lake 345.0 kV
Loading: 108.5% (Rating: 231 MVA)
DFAX: 5.1%
Pre-project loading: 102.3%
22. Facility: McClain Energy Facility-Vaseaux Lake 138.0 kV
Contingency: Hnton Ap-Lowmoor 345.0 kV
Loading: 106.9% (Rating: 340 MVA)
DFAX: 6.5%
Pre-project loading: 101.6%
23. Facility: Pitchgen-Briar 138.0 kV
Contingency: Manatee-Hinton Ap 345.0 kV
Loading: 106.4% (Rating: 196 MVA)
DFAX: 6.4%
Pre-project loading: 97.4%
24. Facility: Red Hills-Vaseaux Lake 138.0 kV
Contingency: Lowmoor-Hinton Ap 345.0 kV
Loading: 105.3% (Rating: 343 MVA)
DFAX: 6.5%
Pre-project loading: 100.0%
25. Facility: Coronet-Westwood Generation Llc 500.0 kV Line
Contingency: Aadia-Montour Staton 500.0 kV Line
Loading: 104.8% (Rating: 2,444 MVA)
DFAX: 8.4%
Pre-project loading: 103.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ioco-Kean Mtn And Grassy Crk 138.0 kV	Rebuild line with higher-capacity conductor	\$5,000,000	15% (\$737,000)
Kean Mtn And Grassy Creek 345/1 kV Autotransformer	Replace transformer with higher-rated unit	\$15,244,000	12% (\$1,793,000)
Kean Mtn And Grassy Crk Substation 138/1 kV Transformer	Replace transformer with higher-rated unit	\$4,457,000	12% (\$524,000)
Tloga-Parke 138.0 kV line	Rebuild line with higher-capacity conductor	\$14,228,000	16% (\$2,232,000)
Scott Branch-Sidney Coal 138.0 kV	Rebuild line with higher-capacity conductor	\$9,122,000	50% (\$4,602,000)
Wansley Combined Cycle-Massapequa 138.0 kV L	Rebuild line with higher-capacity conductor	\$4,576,000	48% (\$2,199,000)
Leesburg Picciola-Orovada 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,880,000	13% (\$517,000)
Hinton Ap-Antelope Vly Station 138.0 kV	Rebuild line with higher-capacity conductor	\$16,518,000	6% (\$1,057,000)
Parke 138/1 kV Autotransformer	Replace transformer with higher-rated unit	\$4,585,000	16% (\$719,000)
Stock Island-Sta 11 138.0 kV	Rebuild line with higher-capacity conductor	\$4,738,000	13% (\$627,000)
Springbank-Sta11 138.0 kV	Rebuild line with higher-capacity conductor	\$8,854,000	13% (\$1,172,000)
Optical Coating Lab-Minnix Mountain 138.0 kV	Rebuild line with higher-capacity conductor	\$3,631,000	16% (\$588,000)
Lovely-Central Ephrata 138.0 kV	Rebuild line with higher-capacity conductor	\$6,200,000	36% (\$2,204,000)
Recway-Minnix Mountain 138.0 kV line	Rebuild line with higher-capacity conductor	\$7,071,000	16% (\$1,144,000)
Recway-Prairie City 138.0 kV	Rebuild line with higher-capacity conductor	\$1,856,000	16% (\$300,000)
Tortilla-Lovely 138.0 kV	Rebuild line with higher-capacity conductor	\$7,310,000	36% (\$2,598,000)
Sidney Coal-Bandon 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,713,000	36% (\$2,031,000)
Bandon-Tortilla 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,558,000	36% (\$1,265,000)
Lonesome Pine-Municipal Light 138.0 kV	Reconductor line	\$2,725,000	16% (\$441,000)
Municipal Light-Daffin Gin 138.0 kV Line	Reconductor line	\$3,517,000	16% (\$569,000)
Lonesome Pine-Prairie City 138.0 kV Line	Reconductor line	\$1,031,000	16% (\$167,000)
Mcclain Energy Facility-Vaseaux Lake 138.0 kV	Reconductor line	\$4,170,000	11% (\$477,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Briar-Pitchgen 138.0 kV	Reconductor line	\$991,000	14% (\$143,000)
Red Hills-Vaseaux Lake 138.0 kV	Reconductor line	\$4,750,000	11% (\$544,000)
Coronet-Westwood Generation Llc 500.0 kV	Reconductor line	\$32,942,000	5% (\$1,660,000)
	Total allocated to project		\$30,310,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.